Beyond Recognition Accuracy

Efficient CPU Streaming ASR for Dysarthric Speech

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SAPC2 Track 2 (Streaming ASR) | NeurIPS 2026 Workshop

Outline

1. Motivation & Challenge
2. Data
3. Method
4. Results
5. Discussion & Conclusion

Motivation & Challenge

Problem Statement

Dysarthric Speech

What was said?

- Motor speech disorder linked to **Parkinson's, cerebral palsy, ALS, Down syndrome, stroke**.
- Slurred, slow or irregular speech that varies **across speakers** and even **within one utterance**.
- ASR trained on typical voices breaks down.

Real-Time Streaming

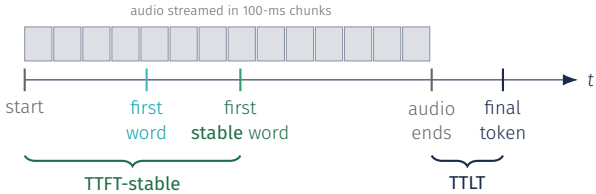
Under what constraints?

- Text must be emitted **incrementally** from short audio chunks.
- **No future context** — the model cannot "wait and listen".
- Must run on **CPU only** with strict latency budgets.

Goal: accessible ASR that is **accurate**, **fast** and **stable** for people with speech impairments.

SAPC2 Track 2 at a Glance

- **Speech Accessibility Project Challenge 2**: shifts focus from offline GPU systems to **real-time** accessibility.
- **Track 2 rules**: audio arrives in **100-ms chunks**; inference strictly on **CPU**.
- **Judged on both**:
  - Accuracy — **CER** & **WER**
  - Responsiveness — **TTFT-stable** & **TTLT**



Metric	Meaning
CER / WER	character / word error rate
TTFT-stable	time until first word that *never changes*
TTLT	delay of final token after audio ends

Research Gaps Addressed

01. Offline-Only SOTA

2025 SAP winners used **Parakeet-TDT 1.1B** & **Whisper large-v3**: non-causal, GPU-bound.

02. Accuracy–Latency

How **compact causal transducers** trade accuracy against **CPU latency** is unexplored.

03. Hidden Variation

Clinical **etiology labels** alone do not explain **where** ASR fails.

Raw speed (RTF) $\neq$ user-perceived responsiveness: emission delay, **hypothesis stability** and endpointing matter too.

Our Contributions

1. **Domain-adapted causal Zipformer** (66.1M) for SAPC2 Track 2 with randomized chunk training, SpecAugment, TTS + TORGO augmentation.
2. **Digit-token vocabulary transplantation** (500 $\rightarrow$ 510 BPE) without overwriting pretrained weights.
3. **Mixed-precision INT8 ONNX** deployment via sherpa-onnx: **22.18% CER / 31.73% WER** on official Test1 (baseline 34.59% CER).
4. **Perceptual error stratification** over 437+ manually annotated utterances: clarity drives errors at least as much as etiology.
5. **Cadrate ASR**: a 2.70M-param dual-rate CNN–GRU prototype probing the lower limit of CPU compute.

Data

Dataset Composition

Split / Source	Utts	Hours	Spk.	Key characteristics
SAPC2 Train	336,075	742.72	875	PD 311, CP 167, ALS 167, DS 145, Stroke 85
SAPC2 Dev	47,929	106.62	124	0% speaker overlap; 20.02% prompt overlap
Synthetic TTS (Qwen + Kokoro)	41,850	—	—	timing, channel, volume perturbations
TORGO + 6-stage aug.	suppl.	—	—	noise, stretch, pitch, LPF, echo, reverb

- **Zero speaker overlap** ⇒ strict zero-shot speaker evaluation.
- 82% of recordings are **under 10 s**.
- ALS longest (8.97 s avg), Down syndrome shortest (6.73 s).
- 68% of files share **voice-assistant commands** (“Hey Google”, “Alexa”, … each 2,100+ times).
- Trained on **disfluency-free normalized** transcripts: **88,694** unique targets.

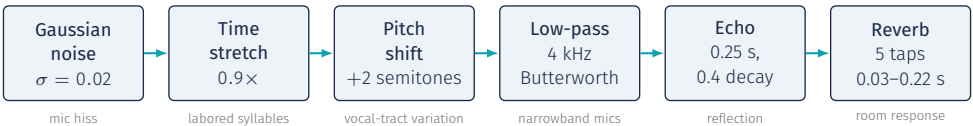
Augmentation: Synthetic Speech + Acoustic Hardening

Lexical diversity

- **41,850** TTS clips from **Qwen-TTS** & **Kokoro-TTS**
- **TORGO** dysarthric recordings

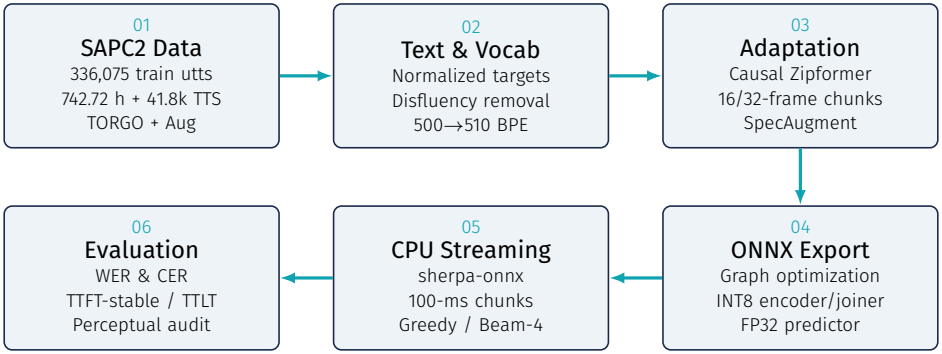
Acoustic diversity: 6-stage distortion pipeline

Separates **channel degradation** from **dysarthric articulation**.



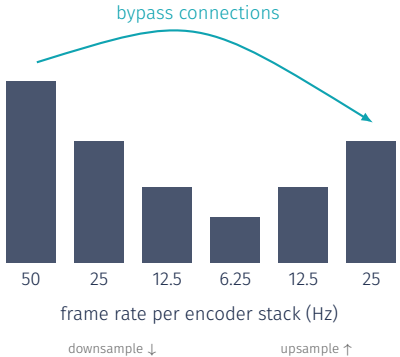
Method

System Pipeline Overview



Primary System: Causal Zipformer Transducer

- Pretrained streaming **Zipformer RNN-T**, ∼**66.1M** params (Icefall).
- **Multi-rate causal encoder** on 80-dim log-Mel fbank — **no future lookahead**.
- **Stateless predictor** + **pruned transducer loss**.
- Mid stacks run at low frame rate ⇒ cheap long context.



Why Zipformer? A **compact, causal** encoder that already streams — adaptation, not size, is the lever.

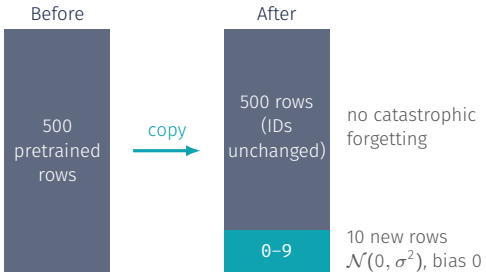
Vocabulary Transplantation: 500 → 510 Tokens

The Problem

Normalized SAPC2 transcripts contain **numerals**; the base 500-token BPE splits them **sub-optimally**.

The Solution

- Append 10 ASCII digits **0–9**.
- Original **500 embedding + joiner rows kept intact**.
- New rows: **scaled Gaussian** init, **zero bias**.

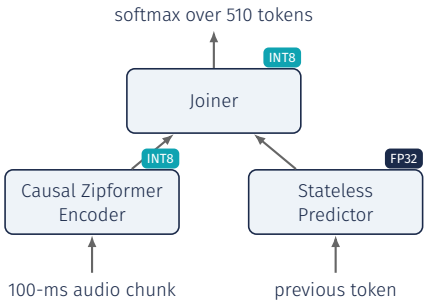


Fine-Tuning Setup

Setting	Value
Hardware	1× RTX 4070 (12 GB)
Precision	FP16 mixed
Base learning rate	3×10^{-5}
Batching	dynamic, $\leq$120 s audio
Augmentation	SpecAugment
Chunk size	random 16–32 frames
Left context	128 or 256 frames
Epochs	19 (E19 selected)

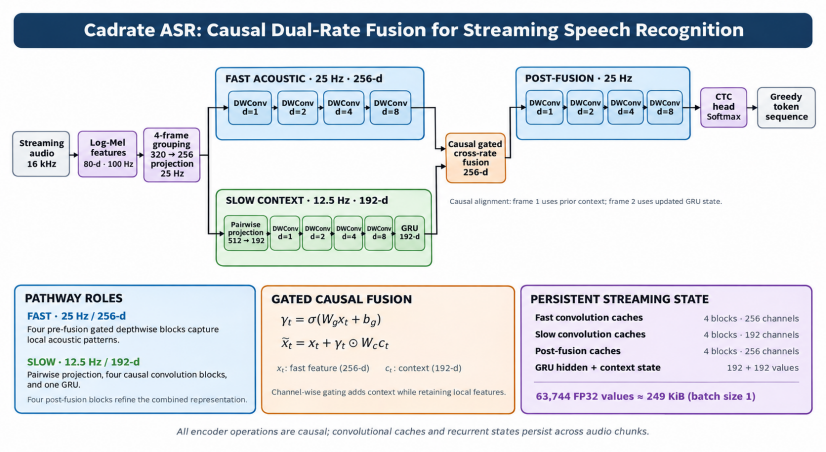
- **Randomized causal chunks** match the 100-ms streaming harness at test time.
- Validation loss & **streaming WER** tracked every epoch.
- Single consumer GPU — the full recipe is **low-budget**.

CPU Deployment: Mixed-Precision INT8 ONNX



- E19 checkpoint → **ONNX**, graph-optimized.
- **INT8 dynamic quantization** on encoder & joiner (heavy compute).
- Lightweight predictor kept in **FP32**.
- Served with **sherpa-onnx**, 100-ms chunks.
- Decoding: **greedy** (official) and **4-path modified beam** with 0.96–1.28 s tail padding.

Prototype: Cadrate ASR (2.70M params)



25 Hz fast acoustic path (8 gated depthwise dilated conv blocks, k=7, $d \in \{1, 2, 4, 8\}$) + **12.5 Hz slow** context path (4 conv blocks + GRU), joined by **causal sigmoid-gated fusion** → CTC.

Zipformer vs. Cadrate ASR

Specification	Adapted Zipformer (Primary)	Cadrate ASR (Prototype)
Encoder	Multi-rate causal Transformer (6 stacks)	Dual-rate causal depthwise CNN–GRU
Temporal rates	50, 25, 12.5, 6.25, 12.5, 25 Hz	25 Hz fast & 12.5 Hz slow
Cross-rate	down/upsampling, bypasses	causal gated additive fusion
Objective	Pruned RNN-T (greedy / beam-4)	Framewise CTC (greedy)
Inference state	attention caches + RNN-T predictor	**fixed 249 KiB** conv/GRU caches
Parameters	∼66.1M (pretrained + fine-tuned)	∼2.70M (from scratch)

$$N_{\text{state}} = 2(256)(6)(1{+}2{+}4{+}8) + (192)(6)(1{+}2{+}4{+}8) + 192 + 192 = 63{,}744 \text{ values} \approx \textbf{249 KiB (FP32)}$$

Time complexity is **linear** in sequence length — no growing attention cache.

Results

Official SAPC2 Test1 Leaderboard

System (CPU streaming, 100-ms chunks)	WER (%) ↓	CER (%) ↓	TTFT-stable ↓	TTLT ↓
SAPC2 Track 2 Baseline	52.77	34.59	—	367 ms
Ours: Zipformer INT8 (Greedy)	**31.73**	**22.18**	**1.38 s**	**117 ms**

−12.41
absolute CER points

−21.04
absolute WER points

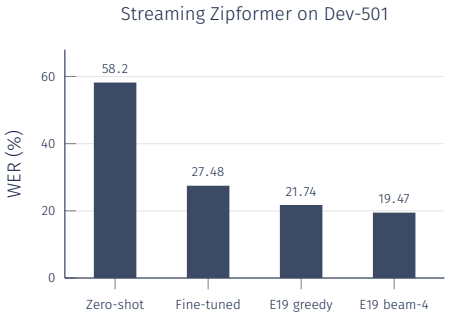
3.1×
faster finalization (TTLT)

Final Outcome

CER 34.59% → **22.18%** with a 117 ms median last-token delay on CPU.

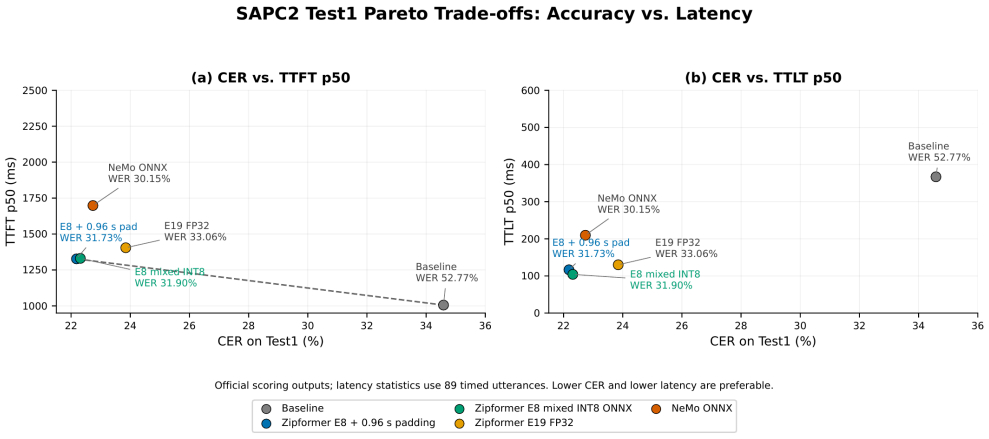
Streaming vs. Offline Reality

System	Split	WER	CER
Whisper Large-v3 (0-shot)	Dev offline	36.56	15.34
Parakeet RNN-T (FT)	Dev offline	22.22	9.67
↑ both RTF > 1.0 on CPU — fail the streaming harness			
Zipformer pretrained (0-shot)	Dev-501	58.20	34.91
Zipformer fine-tuned (val)	Dev-501	27.48	18.90
E19 INT8 greedy	Dev-501	21.74	—
E19 INT8 beam-4	Dev-501	**19.47**	**12.59**



Domain adaptation cut streaming WER by ∼**3**× — and beam-4 adds a further gain **without retraining**.

Accuracy–Latency Trade-offs on Test1



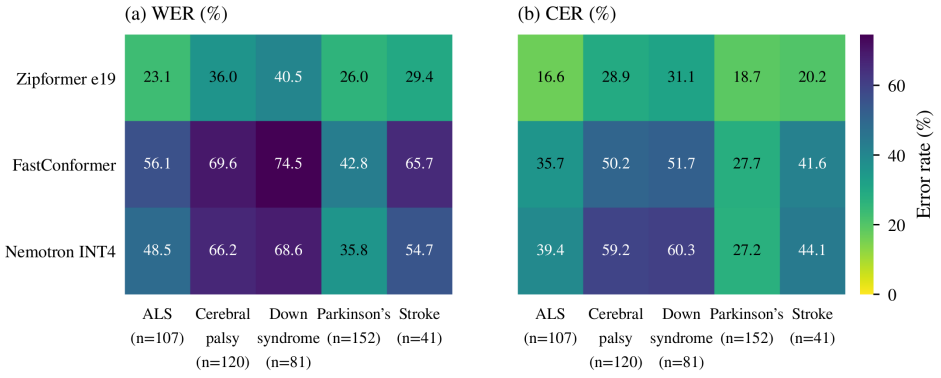
Official scoring outputs; latency over 89 timed utterances. Lower-left is better. Our INT8 variants sit at the accuracy frontier with **3× lower TTLT** than the baseline.

Decoding Ablation (15 configurations, Dev-100)

Configuration	WER	CER	Time (s)
INT8, beam 4, pad 1.28 s	**19.44**	**12.58**	277
INT8, beam 4	19.47	12.59	**157**
INT8, beam 6	19.53	12.72	1086
FP32, beam 4	19.64	12.82	457
INT8, beam 12	19.73	12.88	1496
INT8, beam 4, blank 0.75	20.27	13.43	204
FP32, greedy	21.83	14.53	269
INT8, greedy	22.02	14.70	140
FP32, greedy, blank 2.0	24.83	17.72	513

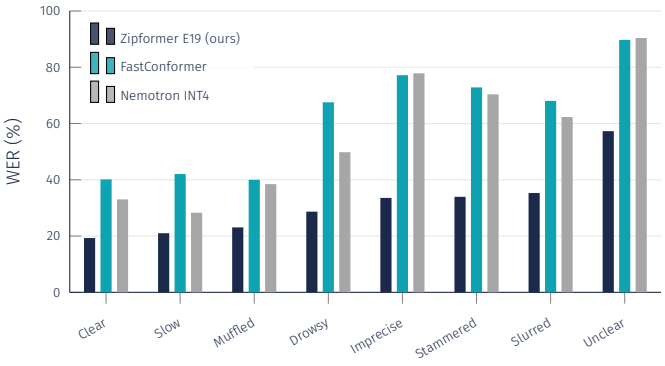
- Greedy $\rightarrow$ **beam-4**: -2.55 WER for only **+17 s** decode time.
- **INT8 beam-4 is 2.9$\times$ faster** than FP32 beam-4 — and slightly more accurate.
- **Wider beams** (6, 12) cost 7–10$\times$ time for no gain.
- **1.28 s tail padding** avoids truncating prolonged dysarthric syllables.
- Blank penalties hurt.

Errors Across Clinical Etiologies



Parkinson's & ALS: stable acoustics, CER $< 20\%$. **Cerebral palsy & Down syndrome**: severe articulatory/timing variability, error rates 30–47%.

Perceptual Error Stratification



- 437+ utterances **manually audited**: clarity, articulation, fluency, rate.
- Ours wins **every category** by 15–44 abs. WER.
- **Clear** (n=152): 19.30%
  Unclear (n=46): **57.28%**

Perceived vocal **clarity** tracks ASR errors at least as strongly as **clinical etiology**.

Fast First Token $\neq$ Usable Output

Off-the-shelf (57 utts)	First out	Stable	TTLT
Moonshine Tiny	**62**	2986	671
Moonshine Small	124	4252	3325
Moonshine ONNX	2336	2736	131
Off-the-shelf Zipformer	2495	2906	546

All values in ms, referenced to audio-send start.

- Without SAPC2 adaptation every model scores **67.78–74.71% WER**.
- Moonshine emits a token in **62 ms**... but the first **stable** word only after **∼3–4 s**.
- Hypotheses that keep **flickering** are not useful to a listener.

This is why we report **TTFT-stable**,
not just first-token latency or RTF.

Cadrate ASR: The Efficiency Limit

What worked

- **Lower CPU RTF** than Zipformer
- **Lower P95 chunk latency**
- Fixed **249 KiB** state, 25× fewer params

What failed

- Trained **from scratch** on SAPC2 only
- **High WER**; did not generalize to **unseen speakers**

Low compute is only useful once accuracy crosses a practical threshold — ultra-fast architectures still need **large-scale pretraining**.

Discussion & Conclusion

Four Practical Lessons

01. Adaptation > Size

A 66.1M causal model, **domain-adapted**, beat every unadapted streaming baseline.

02. Accuracy vs. Stability

Beam search lowers WER but **revises partials**; a transcript that **stays put** helps users more.

03. Speed Needs Pretraining

INT8 ONNX streams smoothly; a tiny model from scratch **cannot generalize**.

04. Labels Hide Clarity

WER follows **perceived clarity** (19% vs. 57%) as much as diagnosis.

Limitations

- No per-component ablation (TTS, TORGO, vocab expansion each in isolation).
- Decoding settings tuned on the reported development set.
- **Single-run** point estimates, no confidence intervals.
- Beam-search gains are **local only**, not on the official leaderboard.
- Perceptual tags are **subjective** author annotations, not clinical diagnoses.
- Cadrate ASR is **exploratory**.
- Scope limited to **English** SAPC2 data; SAP data cannot be redistributed.

Conclusion & Future Directions

Summary

Domain-adapted causal Zipformer + digit-token extension + synthetic augmentation, deployed as INT8 ONNX on CPU: **22.18% CER / 31.73% WER** on official Test1 (baseline 34.59% CER), **1.38 s** TTFT-stable, **117 ms** TTLT.

Next Steps

- Robustness on **unclear, slurred, imprecise** speech.
- Reduce **stable-output latency**, not just first-token latency.
- Lightweight architectures that **generalize to unseen speakers** (pretrained Cadrate).
- Standardize **perceptual benchmarks** alongside clinical etiologies.

Accessible streaming ASR must be not only **fast**, but **accurate** and **stable**.

Key References

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SAPC2 data used under the Speech Accessibility Project data-use agreement.

Questions?

Thank you